(spare bedroom temperature). It is important to understand your constraints before purchasing incompatible equipment.

3.2.5 Plan for integration

Before purchasing any sensor, verify it will actually work with your planned infrastructure. Check specific requirements like 2.4GHz vs 5GHz bands, WPA2 vs WPA3 security for WiFi and whether your chosen platform supports the device's specific implementation. Data format mismatches cause significant frustration and extra work. For example, some sensors output raw ADC values requiring complex decoding, while others provide pre-formatted JSON that integrates immediately. My sensors each use different payload formats, requiring custom decoders for every device type, which is manageable for a few sensors but a maintenance nightmare at scale. Some sensors require regular calibration, like my pool pH sensor which must be calibrated monthly using a buffer solution. Understanding these integration requirements before deployment prevents the scenario of having perfectly functional sensors that can't talk to your digital twin platform or produce data in unusable formats.

3.2.6 Choose sensors for a home digital twin

Based on my objectives to reduce energy and water consumption, I need to identify which physical parameters can provide actionable insights. This means thinking beyond obvious measurements like "electricity usage" to understand what specific data enables optimization decisions. This led me to choose the range of sensors shown in table 3.2.